Contribution Journal: Lab Project

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Introduction

This contribution journal serves to document the specific roles and responsibilities of each team member for Module 06 Lab: Regression and Classification Models. The project was a collaborative effort that required a comprehensive machine learning workflow, spanning data preprocessing and model development to evaluation and final documentation. The successful completion of this lab depended on the distinct contributions of Joshua Balderas, Marisabel Morales, and myself. This document outlines our individual efforts to provide a transparent record of our work.

Joshua Balderas: Code Development and Technical Implementation

Joshua served as a primary developer for the technical implementation of our lab workflows in Python. He collaborated extensively on the code infrastructure, importing the essential libraries (pandas, numpy, and sklearn) and writing the scripts to clean and prepare the Titanic dataset. Joshua focused on building the regression pipeline, implementing the LinearRegression model to predict passenger Fare based on Age and Pclass. His responsibilities included splitting the data into training and test sets, training the model, extracting the learned coefficients, and executing test predictions. Furthermore, he computed the evaluation metrics, obtaining a Mean Squared Error of 3364.92 and an RMSE of $58.01[cite: 1], ensuring the technical results were fully executed and visible for our team's analysis.

Marisabel Morales: Co-Development, Classification Pipeline, and Reflective Journal

Marisabel contributed across both technical execution and comprehensive analytical documentation. On the programming side, she worked alongside Joshua to implement the classification workflow, encoding the categorical Sex column into numeric binary values and fitting the LogisticRegression model to predict passenger survival, which achieved a test accuracy of 74.83%[cite: 2]. Beyond script development, Marisabel authored the official Lab Reflective Journal[cite: 1]. Her writing thoroughly synthesized our experimental findings, clearly articulating the foundational differences between predicting continuous numerical values and discrete class labels[cite: 1, 2]. She explained why MSE/RMSE is required for continuous fare estimation while accuracy serves classification, interpreted the role of model coefficients, and detailed how these concepts directly connect to our ongoing coursework and future machine learning projects[cite: 1, 2].

Katherine Delores (Myself): Project Coordination, Repository Administration, and Contribution Journal

My role centered on project coordination, repository management, and core administrative documentation. I organized our team workflow and milestone deadlines, ensuring smooth communication and handoffs between the coding phase and the reflection drafting phase. I maintained our team's GitHub repository structure, verifying that all completed notebook files, data scripts, and submission requirements were properly aligned with course standards. Additionally, my primary written deliverable was compiling and structuring this Contribution Journal, synthesizing each team member's specific technical tasks, metrics, and deliverables to ensure full transparency and accountability in accordance with the lab guidelines.

Conclusion

The successful completion of Module 06 was made possible by the coordinated distribution of responsibilities across our team. Joshua and Marisabel delivered the programming, model fitting, and baseline evaluations across both linear regression and logistic regression[cite: 1, 2], Marisabel synthesized the theoretical mechanics into our reflective journal[cite: 1], and I managed the timeline, repository organization, and contribution tracking. Dividing the work according to our core deliverables allowed us to fulfill all lab objectives efficiently while reinforcing our collective understanding of supervised learning.